

Project 1:

P1. RNA-Seq Thesis Pipeline: what you deliver

1. Methodological work (the thinking)

Check the design: groups, replicates, batch, confounders (age, sex, sequencing run)

Pick the right model: DESeq2 or edgeR for raw counts, limma-voom for large cohorts

Set justified cut-offs: low-count filter, $FDR < 0.05$, $|\log_2FC| > 1$, with a reason for each

Check sample quality: PCA, sample correlation, outlier decision

Choose the enrichment background properly (only expressed genes, not the whole genome)

2. Technical work (the doing)

Pull GEO data (GEOparse or the supplementary count file) and clean the metadata

Gene ID mapping (Ensembl to symbol) with duplicate handling

Normalization (size factors, VST/rlog) for plots; raw counts for testing

DEG analysis in Python (PyDESeq2) or R (DESeq2)

GO/KEGG enrichment (gseapy or clusterProfiler)

PPI network from STRING, then hub genes with degree, betweenness and MCC (networkx)

Reproducible notebook, fixed seeds, saved tables, session info

3. Analytical work (the biology story)

Up and down gene lists, with the top 10 each explained

Enriched pathways grouped into 3 to 4 themes (for example immune response, cell cycle, metabolism)

Hub genes checked against literature: are they known drivers, or new?

A one-paragraph storyline linking genes, pathways and hubs to the thesis hypothesis

Limitations: small n, batch effect, no wet-lab validation, and suggested qPCR targets

4. Report and visualization work (the polish)

PCA plot and sample heatmap (QC figure)

Volcano plot with labelled top genes

Clustered heatmap of top DEGs, with annotation bars for group and batch

GO/KEGG dot plot and bar plot (clean colours, sorted)

PPI network with hub genes highlighted by size and colour

One consistent theme: journal-ready fonts, colour-blind-safe palette, 300 DPI PNG and SVG

Numbered figure legends and table captions

5. Written text

Methods: tools, versions, thresholds, in thesis style

Results: 4 to 5 paragraphs that follow the figures in order

Short discussion points: what the biology means

Where you act as proxy or skill-gap filler

Student's gap

What you do

"I only know one tool"

Run the full pipeline and explain each step

"I can't explain the biology"

Write the pathway and hub-gene story

"My figures look basic"

Redo them in a professional, consistent style

"I can't write methods"

Draft methods and results from the actual analysis

"Supervisor asks why this cutoff"

Give a short justification note for each choice

Selling line for the portfolio: "You give me the count table. I give you the story, the figures and the chapter."

Project 2:

Thesis-to-Manuscript Builder: what you deliver

1. Methodological work (the thinking)

Find the one core story: a thesis has many findings, but a paper needs one clear message

Choose the target journal first: scope, word limit, IMRaD rules, reference style, open access or not

Decide what to keep, merge or drop (extra aims, long background, negative side results)

Map each thesis chapter to an IMRaD section

Check reporting guidelines (CONSORT, STROBE, PRISMA, ARRIVE) and add the checklist

Plan authorship, ethics statement and data availability text

2. Technical work (the doing)

Parse the thesis file (Word or PDF) into sections with python-docx or PyMuPDF

Rank key sentences with a summarizer and keyword scoring (TF-IDF or sentence embeddings)

Word-budget tool: Intro 500, Methods 800, Results 1,200, Discussion 1,000

Rebuild figures and tables at journal size (300 to 600 DPI, fixed fonts, single or double column)

Reference cleaner: DOI lookup, remove duplicates, convert to the journal style (Vancouver, APA, etc.)

Readability scores (Flesch, grammar and passive-voice check)

Similarity check against the thesis itself and public text, so self-plagiarism is flagged before submission

Export a formatted .docx with title page, abstract, headings, numbered figures and tables

3. Analytical work (the editing judgment)

Which results answer the main question, and which go to supplementary files

Tighten the abstract: Background, Methods, Results, Conclusion, with exact numbers

Rewrite the discussion: 3 main findings, comparison with literature, limitations, one clear conclusion

Fix claims that go beyond the data (a common reason for reviewer rejection)

Check that numbers match across abstract, text, tables and figures

Suggest 3 backup journals in case of rejection

4. Report and visualization work (the polish)

Before/after word-count bar chart per section

Merged multi-panel figures (A, B, C) with consistent colours

Clean, three-line style tables

Readability and similarity summary page for the client

Submission checklist: title page, highlights, keywords, figure files, ethics, conflict of interest

Cover letter: why the work matters, why this journal, no duplicate submission

5. Written text

Full IMRaD draft in journal language

Abstract and keywords

Cover letter

Short response-to-reviewers template for later

Where you act as proxy or skill-gap filler

Researcher's gap

What you do

"I can't cut 200 pages to 4,000 words"

Select, compress and rewrite section by section

"My English gets flagged"

Improve language and readability

"Journal rejected my format"

Match the exact author guidelines

"My figures look like thesis figures"

Redo them in journal style

"I'm afraid of plagiarism flags"

Run the similarity check and rewrite overlaps

"I don't know how to write a cover letter"

Write one tailored to the journal

Selling line for the portfolio: "You give me the thesis. I give you a submission-ready paper, in weeks, not months."